

A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-2800905})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 27 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K , the normalized generator σ_x , the pair (a', J) representing the torsion class e_j , the auxiliary divisor I' , the compactly represented element t , a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1 - \sigma_x)^2 I' + \text{div}(t) + i_x(J) = 0, \quad N_x(t) = a'.$$

Facts marked *checked* are recomputed exactly by the generating script; the full verification is performed by `verify_certificate.c` as described in the README of this directory.

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 + 2800905 = 0$, of discriminant -11203620 . Class group invariants $(10 \ 10 \ 10)$, class number 1000; the three stored generator ideals (HNF) are $\begin{pmatrix} 426 & 51 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 1255 & 755 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 13 & 7 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a , column e_1

Prescribed character vector $(1 \ 0 \ 0)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 - 1027y^3 + (8s - 33869)y^2 + (340s - 648684)y + (2688s - 3346560)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 2053y^8 - 65684y^7 - 174901y^6 + 64171174y^5 + 26 \\ & 65457137y^4 + 66051314232y^3 + 1091726153376y^2 + 9461326049280y \\ & + 31436965969920 \end{aligned}$$

Coordinates of σ_x in the absolute integral basis: $(0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + \left(\frac{403}{16956588762216} \right) s, \quad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. *Checked:* $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $13070819502 = 2 \cdot 3^4 \cdot 13 \cdot 179 \cdot 34673$. The element t is stored in compact form as a product of 130 factors with signed exponents of absolute value up to 61432318; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 2: character a , column e_2

Prescribed character vector $(1 \ 0 \ 0)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 - 1027y^3 + (8s - 33869)y^2 + (340s - 648684)y + (2688s - 3346560)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 2053y^8 - 65684y^7 - 174901y^6 + 64171174y^5 + 26 \\ & 65457137y^4 + 66051314232y^3 + 1091726153376y^2 + 9461326049280y \\ & + 31436965969920 \end{aligned}$$

Coordinates of σ_x in the absolute integral basis: $(0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + \left(\frac{16828}{2498705294406001} \right) s, \quad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. *Checked:* $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $10587476640 = 2^5 \cdot 3 \cdot 5 \cdot 13 \cdot 1696711$. The element t is stored in compact form as a product of 128 factors with signed exponents of absolute value up to 27582645; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 3 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 3: character a , column e_3

Prescribed character vector $(1 \ 0 \ 0)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 - 1027y^3 + (8s - 33869)y^2 + (340s - 648684)y + (2688s - 3346560)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 - 2053y^8 - 65684y^7 - 174901y^6 + 64171174y^5 + 2665457137y^4 + 66051314232y^3 + 1091726153376y^2 + 9461326049280y + 31436965969920$$

Coordinates of σ_x in the absolute integral basis: $(0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + \left(\frac{191}{137858491849}\right)s, \quad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. *Checked:* $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $2099627642060 = 2^2 \cdot 5 \cdot 13 \cdot 2789 \cdot 2895479$. The element t is stored in compact form as a product of 133 factors with signed exponents of absolute value up to 63178993; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 1 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 4: character b , column e_1

Prescribed character vector $(0 \ 1 \ 0)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 680y^3 + 8sy^2 - 115836y - 240s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 1360y^8 + 230728y^6 + 21720960y^4 + 2662503696y^2 + 161332128000$$

Coordinates of σ_x in the absolute integral basis: $(0 \ 0 \ -1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + \left(\frac{403}{16956588762216}\right)s, \quad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. *Checked:* $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $412192614796 = 2^2 \cdot 107 \cdot 963066857$. The element t is stored in compact form as a product of 161 factors with signed exponents of absolute value up to 428347; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 5: character b , column e_2

Prescribed character vector $(0 \ 1 \ 0)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 680y^3 + 8sy^2 - 115836y - 240s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 1360y^8 + 230728y^6 + 21720960y^4 + 2662503696y^2 + 161332128000$$

Coordinates of σ_x in the absolute integral basis: $(0 \ 0 \ -1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + \left(\frac{16828}{2498705294406001}\right)s, \quad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. *Checked:* $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $247528120320 = 2^{13} \cdot 3^5 \cdot 5 \cdot 13 \cdot 1913$. The element t is stored in compact form as a product of 159 factors with signed exponents of absolute value up to 18128; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 6: character b , column e_3

Prescribed character vector $(0 \ 1 \ 0)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 680y^3 + 8sy^2 - 115836y - 240s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 1360y^8 + 230728y^6 + 21720960y^4 + 2662503696y^2 + 161332128000$$

Coordinates of σ_x in the absolute integral basis: (0 0 -1 0 1 0 0 0 0 0).

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + \left(\frac{191}{137858491849}\right)s, \quad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. *Checked:* $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $1633673869360 = 2^4 \cdot 5 \cdot 13 \cdot 67^2 \cdot 349931$. The element t is stored in compact form as a product of 164 factors with signed exponents of absolute value up to 1065502; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (2 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 7: character c , column e_1

Prescribed character vector (0 0 1).

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s - 123)y^3 + (10s + 24508)y^2 + (14s - 189214)y + (-240s + 416904)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 - 245y^8 + 49262y^7 + 2388590y^6 - 60834832y^5 + 848020060y^4 - 7248384008y^3 + 43341537640y^2 - 176590228512y + 335141073216$$

Coordinates of σ_x in the absolute integral basis: (0 0 0 -1 0 0 0 1 1 0).

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + \left(\frac{403}{16956588762216}\right)s, \quad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. *Checked:* $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $244984828968 = 2^3 \cdot 3^3 \cdot 11^3 \cdot 113 \cdot 7541$. The element t is stored in compact form as a product of 113 factors with signed exponents of absolute value up to 30412; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 8: character c , column e_2

Prescribed character vector $(0 \ 0 \ 1)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s - 123)y^3 + (10s + 24508)y^2 + (14s - 189214)y + (-240s + 416904)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 245y^8 + 49262y^7 + 2388590y^6 - 60834832y^5 + 84 \\ & 8020060y^4 - 7248384008y^3 + 43341537640y^2 - 176590228512y + 33 \\ & 5141073216 \end{aligned}$$

Coordinates of σ_x in the absolute integral basis: $(0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + \left(\frac{16828}{2498705294406001} \right) s, \quad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. *Checked:* $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $7544716245 = 3^6 \cdot 5 \cdot 11 \cdot 188171$. The element t is stored in compact form as a product of 116 factors with signed exponents of absolute value up to 113726; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 9: character c , column e_3

Prescribed character vector $(0 \ 0 \ 1)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s - 123)y^3 + (10s + 24508)y^2 + (14s - 189214)y + (-240s + 416904)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 245y^8 + 49262y^7 + 2388590y^6 - 60834832y^5 + 84 \\ & 8020060y^4 - 7248384008y^3 + 43341537640y^2 - 176590228512y + 33 \\ & 5141073216 \end{aligned}$$

Coordinates of σ_x in the absolute integral basis: $(0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + \left(\frac{191}{137858491849}\right)s, \quad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. *Checked:* $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $583425695160 = 2^3 \cdot 3^4 \cdot 5 \cdot 7^2 \cdot 11^4 \cdot 251$. The element t is stored in compact form as a product of 106 factors with signed exponents of absolute value up to 8752; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 1 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \ 1 \ 0)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (s - 62)y^3 + (-7s - 11955)y^2 + (6s - 75342)y + (1128s - 626400)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 120y^8 - 23662y^7 + 2701885y^6 - 38681682y^5 + 32 \\ & 5625238y^4 + 7962666480y^3 - 23477418216y^2 + 132301507680y + 39 \\ & 56203667520 \end{aligned}$$

Coordinates of σ_x in the absolute integral basis: $(1 \ 0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + \left(\frac{403}{16956588762216}\right)s, \quad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. *Checked:* $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $340301732544 = 2^6 \cdot 3^3 \cdot 241 \cdot 817153$. The element t is stored in compact form as a product of 124 factors with signed exponents of absolute value up to 1003112; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \ 1 \ 0)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (s - 62)y^3 + (-7s - 11955)y^2 + (6s - 75342)y + (1128s - 626400)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 120y^8 - 23662y^7 + 2701885y^6 - 38681682y^5 + 32 \\ & 5625238y^4 + 7962666480y^3 - 23477418216y^2 + 132301507680y + 39 \\ & 56203667520 \end{aligned}$$

Coordinates of σ_x in the absolute integral basis: $(1 \ 0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + \left(\frac{16828}{2498705294406001} \right) s, \quad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. *Checked:* $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $1655028426480 = 2^4 \cdot 3 \cdot 5 \cdot 11^2 \cdot 13 \cdot 4383949$. The element t is stored in compact form as a product of 124 factors with signed exponents of absolute value up to 919270; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \ 2 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \ 1 \ 0)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (s - 62)y^3 + (-7s - 11955)y^2 + (6s - 75342)y + (1128s - 626400)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 120y^8 - 23662y^7 + 2701885y^6 - 38681682y^5 + 32 \\ & 5625238y^4 + 7962666480y^3 - 23477418216y^2 + 132301507680y + 39 \\ & 56203667520 \end{aligned}$$

Coordinates of σ_x in the absolute integral basis: $(1 \ 0 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + \left(\frac{191}{137858491849}\right)s, \quad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. *Checked:* $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $97407492600 = 2^3 \cdot 3 \cdot 5^2 \cdot 11^2 \cdot 1341701$. The element t is stored in compact form as a product of 130 factors with signed exponents of absolute value up to 1242228; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \ 4 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \ 0 \ 1)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (-s - 121)y^3 + (16s + 3420)y^2 + (-180s - 305964)y + (-432s + 2001024)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 6y^9 - 233y^8 + 7566y^7 + 2183098y^6 - 84618768y^5 + 179 \\ & 9091024y^4 - 16290272448y^3 + 159330584736y^2 - 788885868672y + \\ & 4526813143296 \end{aligned}$$

Coordinates of σ_x in the absolute integral basis: $(0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + \left(\frac{403}{16956588762216}\right)s, \quad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. *Checked:* $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $233867734740 = 2^2 \cdot 3^2 \cdot 5 \cdot 23 \cdot 761 \cdot 74231$. The element t is stored in compact form as a product of 129 factors with signed exponents of absolute value up to 38956; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \ 1 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \ 0 \ 1)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (-s - 121)y^3 + (16s + 3420)y^2 + (-180s - 305964)y + (-432s + 2001024)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 6y^9 - 233y^8 + 7566y^7 + 2183098y^6 - 84618768y^5 + 179 \\ & 9091024y^4 - 16290272448y^3 + 159330584736y^2 - 788885868672y + \\ & 4526813143296 \end{aligned}$$

Coordinates of σ_x in the absolute integral basis: $(0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + \left(\frac{16828}{2498705294406001} \right) s, \quad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. *Checked:* $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $22894558920 = 2^3 \cdot 3^2 \cdot 5 \cdot 17 \cdot 607 \cdot 6163$. The element t is stored in compact form as a product of 124 factors with signed exponents of absolute value up to 243599; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \ 4 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \ 0 \ 1)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 3y^4 + (-s - 121)y^3 + (16s + 3420)y^2 + (-180s - 305964)y + (-432s + 2001024)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 6y^9 - 233y^8 + 7566y^7 + 2183098y^6 - 84618768y^5 + 179 \\ & 9091024y^4 - 16290272448y^3 + 159330584736y^2 - 788885868672y + \\ & 4526813143296 \end{aligned}$$

Coordinates of σ_x in the absolute integral basis: (0 0 -1 0 0 0 0 -1 0 0).

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + \left(\frac{191}{137858491849}\right)s, \quad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. *Checked:* $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $2154434142720 = 2^9 \cdot 3 \cdot 5 \cdot 17 \cdot 1861 \cdot 8867$. The element t is stored in compact form as a product of 129 factors with signed exponents of absolute value up to 100605; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (0 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector (0 1 1).

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 26y^3 - 12sy^2 + 483525y + 2220s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} + 52y^8 + 967726y^6 + 428473620y^4 + 84564207225y^2 + 13803 \\ & 980202000 \end{aligned}$$

Coordinates of σ_x in the absolute integral basis: (0 1 1 -1 -1 0 0 0 0 0).

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + \left(\frac{403}{16956588762216}\right)s, \quad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. *Checked:* $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $178513655400 = 2^3 \cdot 3^2 \cdot 5^2 \cdot 163 \cdot 608431$. The element t is stored in compact form as a product of 157 factors with signed exponents of absolute value up to 124585997; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \ 1 \ 1)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 26y^3 - 12sy^2 + 483525y + 2220s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 52y^8 + 967726y^6 + 428473620y^4 + 84564207225y^2 + 13803980202000$$

Coordinates of σ_x in the absolute integral basis: $(0 \ 1 \ 1 \ -1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + \left(\frac{16828}{2498705294406001} \right) s, \quad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. *Checked:* $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $372996494700 = 2^2 \cdot 3 \cdot 5^2 \cdot 23 \cdot 179 \cdot 301997$. The element t is stored in compact form as a product of 161 factors with signed exponents of absolute value up to 147592620; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \ 1 \ 1)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 26y^3 - 12sy^2 + 483525y + 2220s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 52y^8 + 967726y^6 + 428473620y^4 + 84564207225y^2 + 13803980202000$$

Coordinates of σ_x in the absolute integral basis: (0 1 1 -1 -1 0 0 0 0 0).

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + \left(\frac{191}{137858491849}\right)s, \quad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. *Checked:* $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $55781154432 = 2^7 \cdot 3^2 \cdot 23 \cdot 2105267$. The element t is stored in compact form as a product of 155 factors with signed exponents of absolute value up to 8392999; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (2 \ 2 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 19: character $2a$, column e_1

Prescribed character vector (2 0 0).

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 - 1027y^3 + (-8s - 33869)y^2 + (-340s - 648684)y + (-2688s - 3346560)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 - 2053y^8 - 65684y^7 - 174901y^6 + 64171174y^5 + 2665457137y^4 + 66051314232y^3 + 1091726153376y^2 + 9461326049280y + 31436965969920$$

Coordinates of σ_x in the absolute integral basis: (2 0 -1 0 0 -1 -1 -1 1 0).

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + \left(\frac{403}{16956588762216}\right)s, \quad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. *Checked:* $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $13070819502 = 2 \cdot 3^4 \cdot 13 \cdot 179 \cdot 34673$. The element t is stored in compact form as a product of 128 factors with signed exponents of absolute value up to 150967808; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 20: character $2a$, column e_2

Prescribed character vector $(2 \ 0 \ 0)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 - 1027y^3 + (-8s - 33869)y^2 + (-340s - 648684)y + (-2688s - 3346560)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 2053y^8 - 65684y^7 - 174901y^6 + 64171174y^5 + 26 \\ & 65457137y^4 + 66051314232y^3 + 1091726153376y^2 + 9461326049280y \\ & + 31436965969920 \end{aligned}$$

Coordinates of σ_x in the absolute integral basis: $(2 \ 0 \ -1 \ 0 \ 0 \ -1 \ -1 \ -1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + \left(\frac{16828}{2498705294406001} \right) s, \quad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. *Checked:* $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $379155860616 = 2^3 \cdot 3 \cdot 13 \cdot 499 \cdot 2435357$. The element t is stored in compact form as a product of 134 factors with signed exponents of absolute value up to 266216466; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 2 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 21: character $2a$, column e_3

Prescribed character vector $(2 \ 0 \ 0)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 - 1027y^3 + (-8s - 33869)y^2 + (-340s - 648684)y + (-2688s - 3346560)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 - 2053y^8 - 65684y^7 - 174901y^6 + 64171174y^5 + 2665457137y^4 + 66051314232y^3 + 1091726153376y^2 + 9461326049280y + 31436965969920$$

Coordinates of σ_x in the absolute integral basis: (2 0 -1 0 0 -1 -1 -1 1 0).

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + \left(\frac{191}{137858491849}\right)s, \quad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. *Checked:* $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $158406917760 = 2^7 \cdot 3^6 \cdot 5 \cdot 7^2 \cdot 13^2 \cdot 41$. The element t is stored in compact form as a product of 128 factors with signed exponents of absolute value up to 363594176; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (0 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 22: character $2b$, column e_1

Prescribed character vector (0 2 0).

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 680y^3 - 8sy^2 - 115836y + 240s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 1360y^8 + 230728y^6 + 21720960y^4 + 2662503696y^2 + 161332128000$$

Coordinates of σ_x in the absolute integral basis: (0 1 0 1 -1 0 0 0 0 0).

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + \left(\frac{403}{16956588762216}\right)s, \quad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. *Checked:* $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $5293226700 = 2^2 \cdot 3^2 \cdot 5^2 \cdot 557 \cdot 10559$. The element t is stored in compact form as a product of 158 factors with signed exponents of absolute value up to 1397128; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 0 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 23: character $2b$, column e_2

Prescribed character vector $(0 \ 2 \ 0)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 680y^3 - 8sy^2 - 115836y + 240s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 1360y^8 + 230728y^6 + 21720960y^4 + 2662503696y^2 + 161332128000$$

Coordinates of σ_x in the absolute integral basis: $(0 \ 1 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + \left(\frac{16828}{2498705294406001}\right)s, \quad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. *Checked:* $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $213512823864 = 2^3 \cdot 3^2 \cdot 107 \cdot 509 \cdot 54449$. The element t is stored in compact form as a product of 161 factors with signed exponents of absolute value up to 391; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \ 0 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 24: character $2b$, column e_3

Prescribed character vector $(0 \ 2 \ 0)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 680y^3 - 8sy^2 - 115836y + 240s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 1360y^8 + 230728y^6 + 21720960y^4 + 2662503696y^2 + 161332128000$$

Coordinates of σ_x in the absolute integral basis: $(0 \ 1 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + \left(\frac{191}{137858491849}\right)s, \quad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. *Checked:* $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $85548070400 = 2^9 \cdot 5^2 \cdot 13^2 \cdot 71 \cdot 557$. The element t is stored in compact form as a product of 163 factors with signed exponents of absolute value up to 4221650; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \ 0 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 25: character $2c$, column e_1

Prescribed character vector $(0 \ 0 \ 2)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s - 123)y^3 + (-10s + 24508)y^2 + (-14s - 189214)y + (240s + 416904)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 2y^9 - 245y^8 + 49262y^7 + 2388590y^6 - 60834832y^5 + 848020060y^4 - 7248384008y^3 + 43341537640y^2 - 176590228512y + 335141073216$$

Coordinates of σ_x in the absolute integral basis: $(0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + \left(\frac{403}{16956588762216}\right)s, \quad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. *Checked:* $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $102545175744 = 2^6 \cdot 3^8 \cdot 11 \cdot 149^2$. The element t is stored in compact form as a product of 108 factors with signed exponents of absolute value up to 57768; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \ 1 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 26: character $2c$, column e_2

Prescribed character vector $(0 \ 0 \ 2)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s - 123)y^3 + (-10s + 24508)y^2 + (-14s - 189214)y + (240s + 416904)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 245y^8 + 49262y^7 + 2388590y^6 - 60834832y^5 + 84 \\ & 8020060y^4 - 7248384008y^3 + 43341537640y^2 - 176590228512y + 33 \\ & 5141073216 \end{aligned}$$

Coordinates of σ_x in the absolute integral basis: $(0 \ -1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + \left(\frac{16828}{2498705294406001} \right) s, \quad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. *Checked:* $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $357730716090 = 2 \cdot 3 \cdot 5 \cdot 11 \cdot 71 \cdot 151 \cdot 101113$. The element t is stored in compact form as a product of 117 factors with signed exponents of absolute value up to 49107; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 27: character $2c$, column e_3

Prescribed character vector $(0 \ 0 \ 2)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s - 123)y^3 + (-10s + 24508)y^2 + (-14s - 189214)y + (240s + 416904)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 245y^8 + 49262y^7 + 2388590y^6 - 60834832y^5 + 84 \\ & 8020060y^4 - 7248384008y^3 + 43341537640y^2 - 176590228512y + 33 \\ & 5141073216 \end{aligned}$$

Coordinates of σ_x in the absolute integral basis: (0 -1 0 0 0 0 1 0 0 0).

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + \left(\frac{191}{137858491849}\right)s, \quad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. *Checked:* $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: **pass**.

The auxiliary pair (t, I')

I' is stored as a 10×10 HNF matrix in the absolute model; its absolute norm is $2474463940740 = 2^2 \cdot 3^5 \cdot 5 \cdot 11 \cdot 113 \cdot 157 \cdot 2609$. The element t is stored in compact form as a product of 106 factors with signed exponents of absolute value up to 11365; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (0 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.